

Secure Deployment Documentation – Nilavan Application

Project Overview

This document explains the complete deployment process of a secure production-ready Next.js application on Ubuntu 24.04 using:

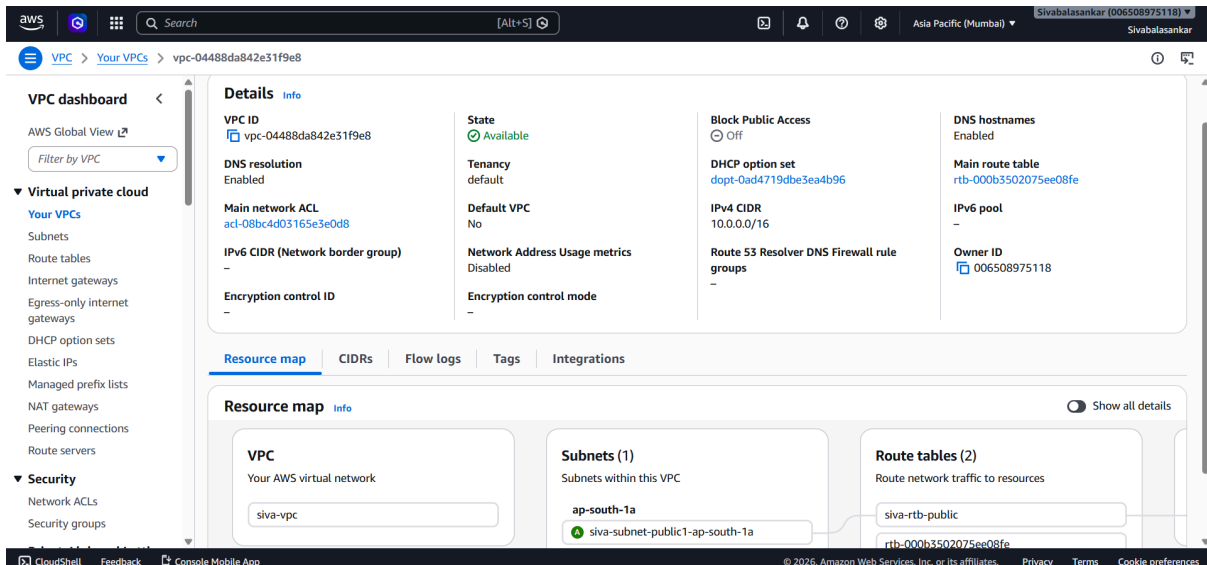
- Ubuntu 24.04 VPS
- Nginx Reverse Proxy
- PM2 Process Manager
- DuckDNS
- SSL using Certbot
- SSH Hardening
- GitHub SSH Deployment
- Elastic IP Configuration

1. Infrastructure Setup

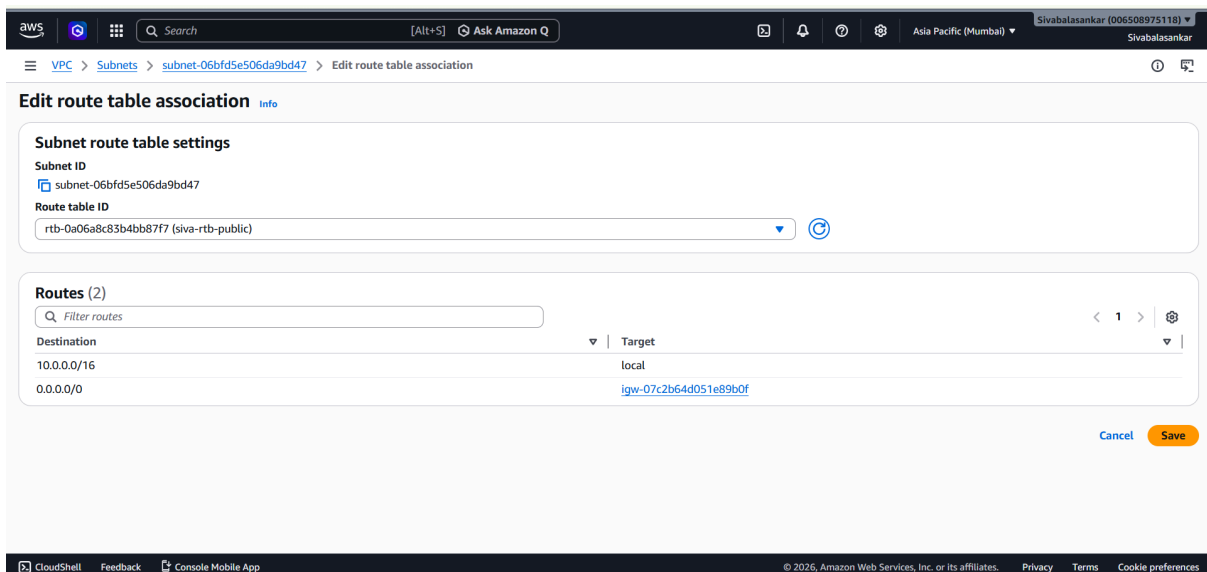
1.1 Created VPC Network

A custom VPC was created with:

- Public subnet
- Internet Gateway (IGW)
- Route table configuration
- Public internet access



- VPC Dashboard
- Subnet Configuration
- Route Table
- Internet Gateway Attached



2. Virtual Machine Creation

2.1 Created Ubuntu VM

VM Details:

Configuration	Value
OS	Ubuntu 24.04 LTS
Instance Type	e2-medium
Public IP	Attached
SSH Access	Enabled

aws

Search

[Alt+S]

Asia Pacific (Mumbai) SivabalanSankar (005508975118) SivabalanSankar

EC2 > Instances

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Instances (1/1) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
Leadtap.ai	i-042b07ee24f20a281	Running	t3.small	3/3 checks passed	View alarms	ap-south-1a	ec2-43-204

i-042b07ee24f20a281 (Leadtap.ai)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary Info

Instance ID

i-042b07ee24f20a281

IPv6 address

-

Hostname type

Private IP address

43.204.69.25

open address

Instance state

Running

Private IP DNS name (IPv4 only)

Private IPv4 addresses

10.0.10.55

Public DNS

ec2-43-204-69-25.ap-south-1.compute.amazonaws.com

open address

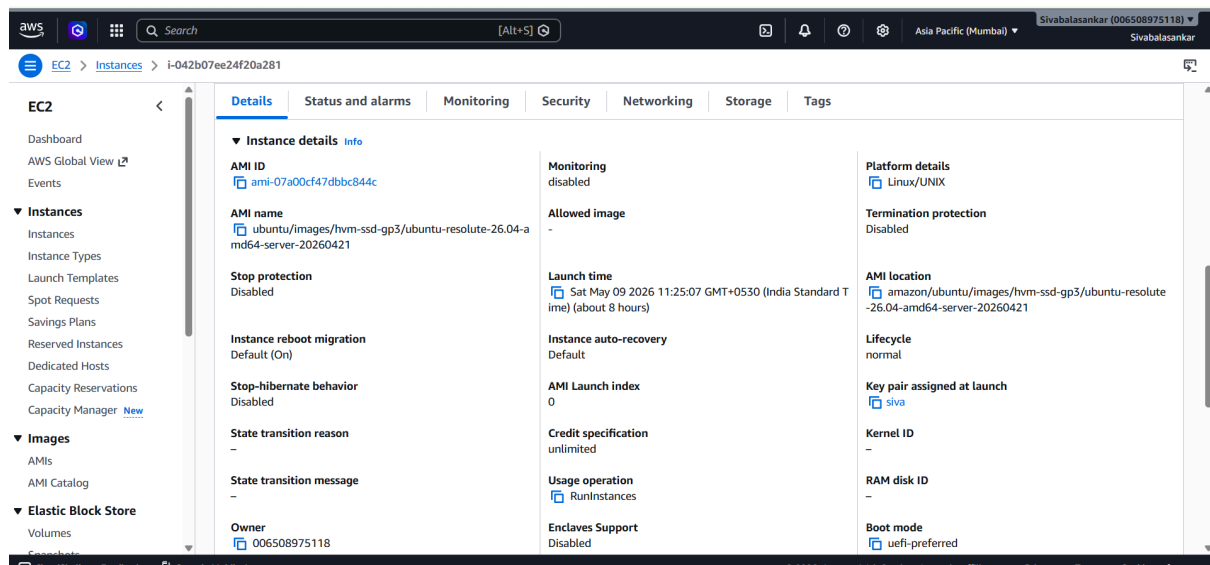
CloudShell

Feedback

Console Mobile App

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- VM instance dashboard
- Running status
- External IP
- Machine type



3. Elastic IP Configuration

3.1 Created and Attached Elastic IP

An Elastic IP was created from the AWS Management Console and attached to the EC2 instance to provide a static public IP address.

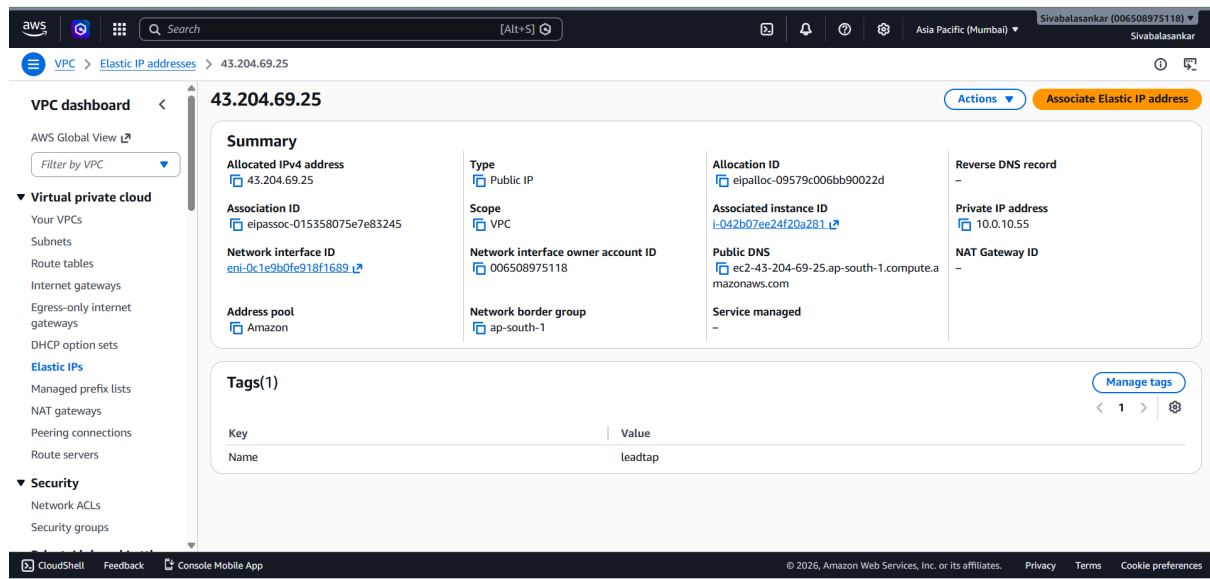
Steps Performed in AWS Portal

1. Logged in to the AWS Management Console.
2. Navigated to:

EC2 Dashboard → Network & Security → Elastic IPs

3. Clicked on: Allocate Elastic IP address
4. Clicked: Allocate
5. Selected the newly created Elastic IP.
6. Clicked: Actions → Associate Elastic IP address
7. Chose:
 - Resource type: Instance
 - Selected the Ubuntu EC2 instance
 - Selected the private IP
8. Clicked: Associate

The Elastic IP was later mapped to the DuckDNS domain.



- Elastic IP allocation page
- Elastic IP attached to EC2 instance
- AWS EC2 dashboard showing Elastic IP association
- Later elastic IP is mapped with DuckDNS

4. User Creation & Hardening

4.1 Created Non-Root User

A dedicated deployment user was created.

Commands Used

```
sudo adduser siva-leadtap  
sudo usermod -aG sudo siva-leadtap
```

```
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mail List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:./nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:./usr/sbin/nologin
dhcpcd:x:996:996:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
messagebus:x:995:995:System Message Bus:/nonexistent:/usr/sbin/nologin
syslog:x:100:101:./nonexistent:/usr/sbin/nologin
systemd-resolve:x:989:989:systemd Resolver:./usr/sbin/nologin
chrony:x:988:988:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
tss:x:987:987:tss user for tpm2:./usr/sbin/nologin
uidd:x:101:103:./run/uidd:/usr/sbin/nologin
sshd:x:986:65534:sshd user:/run/sshd:/usr/sbin/nologin
pollinate:x:102:1:./var/cache/pollinate:/bin/false
tcpdump:x:985:985:tcpdump:/nonexistent:/usr/sbin/nologin
landscape:x:103:106:./var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:984:984:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
polkitd:x:983:983:User for polkitd:./usr/sbin/nologin
ec2-instance-connect:x:104:65534:./nonexistent:/usr/sbin/nologin
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
siva-leadtap:x:1001:1001:siva-leadtap,./home/siva-leadtap:/bin/bash
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$

siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ sudo whoami
[sudo: authenticate] Password:
root
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$
```

5. SSH Security Hardening

5.1 SSH Configuration

SSH was hardened using the following settings:

Setting	Value
SSH Port	2222
Root Login	Disabled
Password Authentication	Disabled
Public Key Authentication	Enabled

SSH Config File

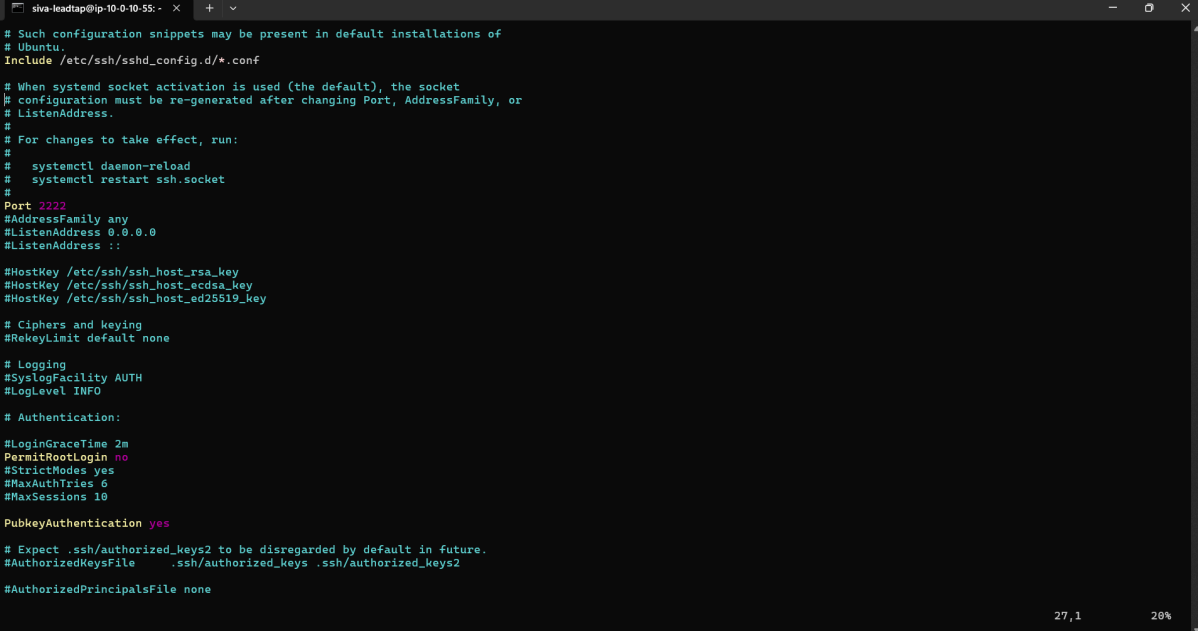
```
sudo nano /etc/ssh/sshd_config
```

Configuration Added

```
Port 2222
PermitRootLogin no
PasswordAuthentication no
PubkeyAuthentication yes
```

Restart SSH

```
sudo systemctl restart ssh
```



```
shiva-leadtap@ip-10-0-10-55: ~$ cat /etc/ssh/sshd_config
# Such configuration snippets may be present in default installations of
# Ubuntu.
Include /etc/ssh/sshd_config.d/*.conf

# When systemd socket activation is used (the default), the socket
# configuration must be re-generated after changing Port, AddressFamily, or
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
#
Port 2222
AddressFamily any
ListenAddress 0.0.0.0
ListenAddress ::

HostKey /etc/ssh/ssh_host_rsa_key
HostKey /etc/ssh/ssh_host_ecdsa_key
HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

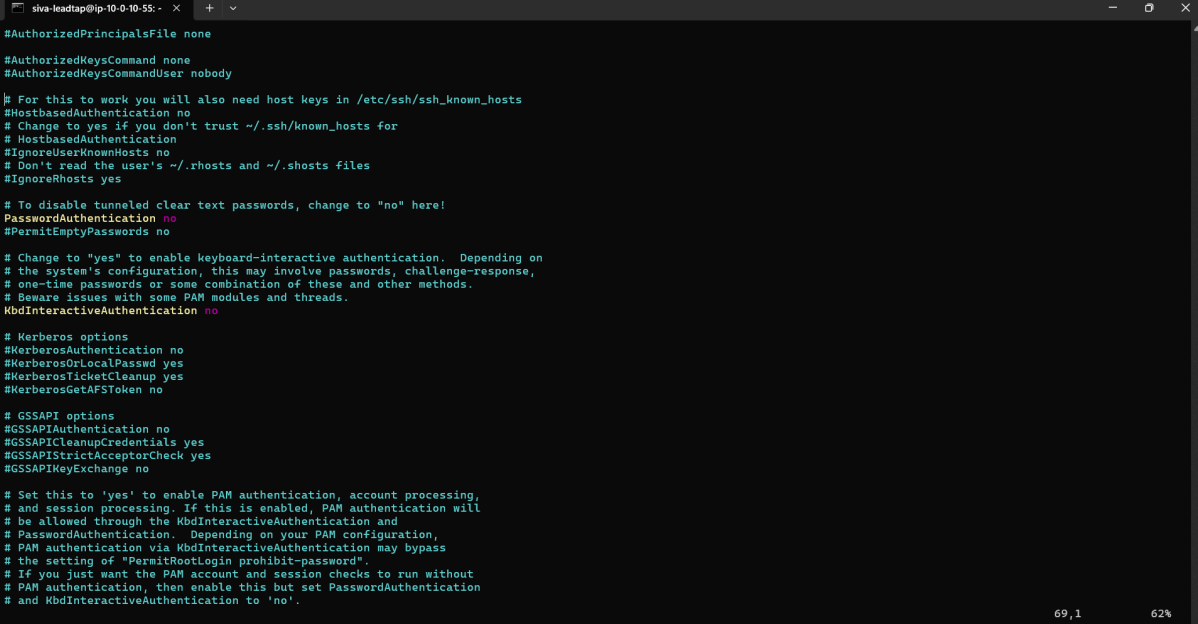
# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
StrictModes yes
MaxAuthTries 6
MaxSessions 10

PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none
```



```
shiva-leadtap@ip-10-0-10-55: ~$ cat /etc/ssh/sshd_config
#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to "no" here!
PasswordAuthentication no
PermitEmptyPasswords no

# Change to "yes" to enable keyboard-interactive authentication. Depending on
# the system's configuration, this may involve passwords, challenge-response,
# one-time passwords or some combination of these and other methods.
# Beware issues with some PAM modules and threads.
KbdInteractiveAuthentication no

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
#KerberosGetAFSToken no

# GSSAPI options
#GSSAPIAuthentication no
#GSSAPICleanupCredentials yes
#GSSAPIStrictAcceptorCheck yes
#GSSAPIKeyExchange no

# Set this to 'yes' to enable PAM authentication, account processing,
# and session processing. If this is enabled, PAM authentication will
# be allowed through the KbdInteractiveAuthentication and
# PasswordAuthentication. Depending on your PAM configuration,
# PAM authentication via KbdInteractiveAuthentication may bypass
# the setting of "PermitRootLogin prohibit-password".
# If you just want the PAM account and session checks to run without
# PAM authentication, then enable this but set PasswordAuthentication
# and KbdInteractiveAuthentication to 'no'.
```

```
sudo systemctl status ssh
```

```
ubuntu@ip-10-0-10-55:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Drop-In: /usr/lib/systemd/system/ssh.service.d
            └─ec2-instance-connect.conf
   Active: active (running) since Sat 2026-05-09 05:55:21 UTC; 22h ago
   Invocation: 225611cbb2934d6fbbd4e62d3c036779
   TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
   Main PID: 992 (sshd)
     Tasks: 1 (limit: 1741)
    Memory: 7.8M (peak: 30.5M)
       CPU: 2.334s
    CGroup: /system.slice/ssh.service
            └─992 "sshd: /usr/sbin/sshd -D -o AuthorizedKeysCommand /usr/share/ec2-instance-connect/eic_run_authorized"

May 09 13:34:47 ip-10-0-10-55 sshd-session[16418]: Accepted publickey for siva-leadtap from 157.50.1.159 port 47118 ssh:
May 09 13:34:47 ip-10-0-10-55 sshd-session[16418]: pam_unix(sshd:session): session opened for user siva-leadtap(uid=1000) by
May 09 14:13:01 ip-10-0-10-55 sshd-session[17246]: Accepted publickey for siva-leadtap from 157.50.7.124 port 36981 ssh:
May 09 14:13:01 ip-10-0-10-55 sshd-session[17246]: pam_unix(sshd:session): session opened for user siva-leadtap(uid=1000) by
May 09 20:14:43 ip-10-0-10-55 sshd-session[19965]: Invalid user from 18.97.19.141 port 34520
May 09 20:14:44 ip-10-0-10-55 sshd-session[19965]: Connection closed by invalid user 18.97.19.141 port 34520 [preauth]
May 10 04:12:24 ip-10-0-10-55 sshd-session[21073]: Accepted publickey for ubuntu from 157.50.1.143 port 46157 ssh2: RSA
May 10 04:12:24 ip-10-0-10-55 sshd-session[21073]: pam_unix(sshd:session): session opened for user ubuntu(uid=1000) by
May 10 04:30:27 ip-10-0-10-55 sshd-session[21378]: Accepted publickey for ubuntu from 157.50.1.143 port 45160 ssh2: RSA
May 10 04:30:27 ip-10-0-10-55 sshd-session[21378]: pam_unix(sshd:session): session opened for user ubuntu(uid=1000) by
lines 1-26/26 (END)
```

ss -tulpn | grep 2222

```
siva-leadtap@ip-10-0-10-55:~$ netstat -tulpn | grep 2222
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)
tcp        0      0 0.0.0.0:2222        0.0.0.0:*           LISTEN     -
tcp6       0      0 :::2222            :::*                 LISTEN     -
siva-leadtap@ip-10-0-10-55:~$
```

sudo cat /etc/ssh/sshd_config

6. SSH Key Authentication

6.1 Added Public Key Authentication

The SSH public key was added to the new deployment user.

Commands Used

```
mkdir -p ~/.ssh
nano ~/.ssh/authorized_keys
chmod 700 ~/.ssh
chmod 600 ~/.ssh/authorized_keys
```

```
siva-leadtap@ip-10-0-10-55:~$ cat ~/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDp3rjggZcgpSI2asMIdcobXJ810ISsslXE74m5wzx+5JvzrWbLvgFzLJyUCZwZcVm+l01/x0GFqK6CFLHjjTHdSnnXe0Xd6fU1KXi4czf0St4g50QumrYB
fXQrFnMR4N028PCwr0MuXzLKA8zvJX3EnRmKlrTiAR4RjwX3qwy1Ch2B1DddOURiitY1KHye6igNoAngaM2/Qgha2X1If7wDR0bnJ7q+AFMGfMAVQRtI1TX6Z0MQiUcI8SfoLyvK30tHBXztQ5YKQbISQR19iH
I1vQst
siva
```

```
siva-leadtap@ip-10-0-10-55:~$ ls -la ~/.ssh/authorized_keys
-rw----- 1 siva-leadtap siva-leadtap 386 May  8 10:26 /home/siva-leadtap/.ssh/authorized_keys
siva-leadtap@ip-10-0-10-55:~$
```


7. Installed Required Software

7.1 Installed Git

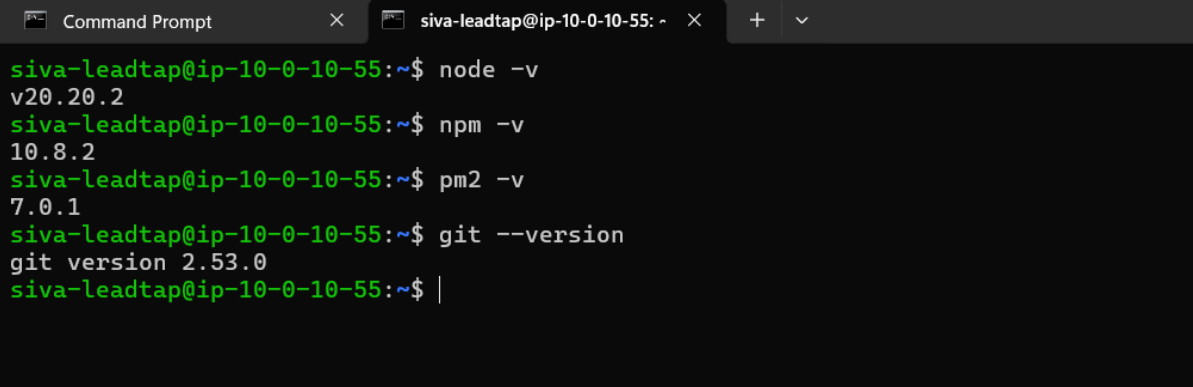
```
sudo apt update  
sudo apt install git -y
```

7.2 Installed Node.js 20

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -  
sudo apt install nodejs -y
```

7.3 Installed PM2

```
sudo npm install -g pm2
```



```
Command Prompt x siva-leadtap@ip-10-0-10-55: ~ x + v  
siva-leadtap@ip-10-0-10-55:~$ node -v  
v20.20.2  
siva-leadtap@ip-10-0-10-55:~$ npm -v  
10.8.2  
siva-leadtap@ip-10-0-10-55:~$ pm2 -v  
7.0.1  
siva-leadtap@ip-10-0-10-55:~$ git --version  
git version 2.53.0  
siva-leadtap@ip-10-0-10-55:~$ |
```

8. Deployment

8.1 Cloned Repository using SSH

Commands Used

```
git clone git@github.com:Leadtap/lt-nilavan.git
```

```

git version 2.35.0
siva-leadtap@ip-10-0-10-55:~$ ssh -T git@github.com
Hi Sivabalasankar2002! You've successfully authenticated, but GitHub does not provide shell access.
siva-leadtap@ip-10-0-10-55:~$ |

siva-leadtap@ip-10-0-10-55:~$ cd lt-nilavan/
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ ls
README.md  components      hooks  next-env.d.ts  package.json  public  tailwind.config.ts
app        components.json lib    package-lock.json  postcss.config.mjs  styles  tsconfig.json
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ |

```

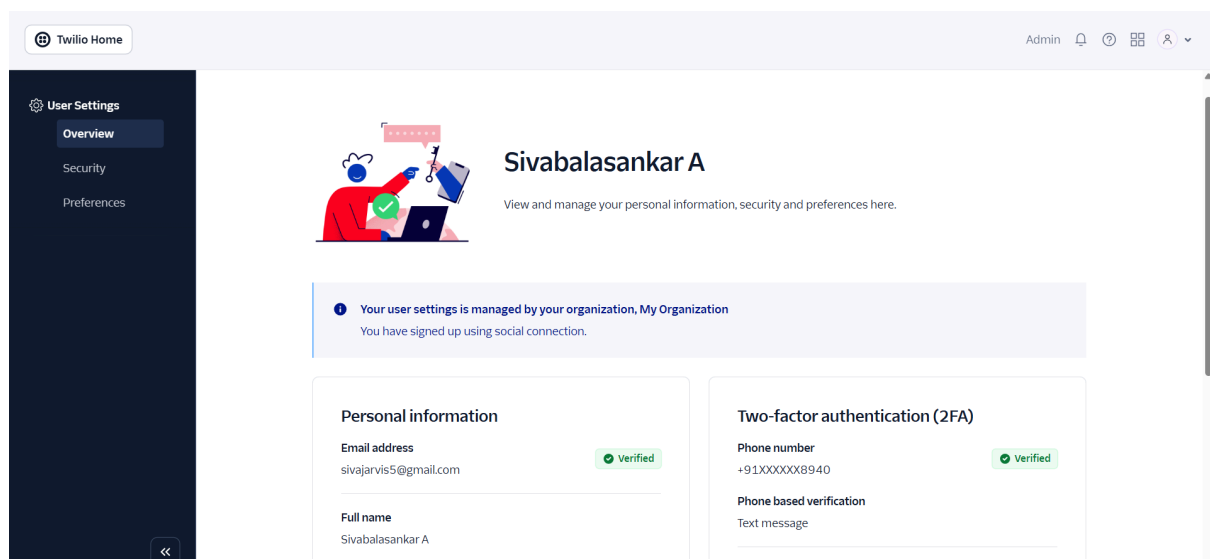
SendGrid Email Service Configuration

Created SendGrid Account and API Key

A SendGrid account was created for handling email functionality from the Next.js application.

The following steps were completed:

Created SendGrid account



Verified sender email address

You are required to include your contact information, including a physical mailing address, inside every promotional email you send in order to comply with the anti-spam laws such as [CAN-SPAM](#) and [CASL](#). You'll find [replacement tags](#) for this information in the footer of all the email designs SendGrid provides. [Learn more](#)

From Name •

Siva Testing



From Email Address •

sivajarvis5@gmail.com



Reply To •

sivajarvis5@gmail.com



Company Address •

22/1 muthu irulappa street, ettayapuram

Company Address Line 2

City •

Coimbatore

Zip Code

635001

Country •

India



Nickname •

Sivabalasankar



Cancel

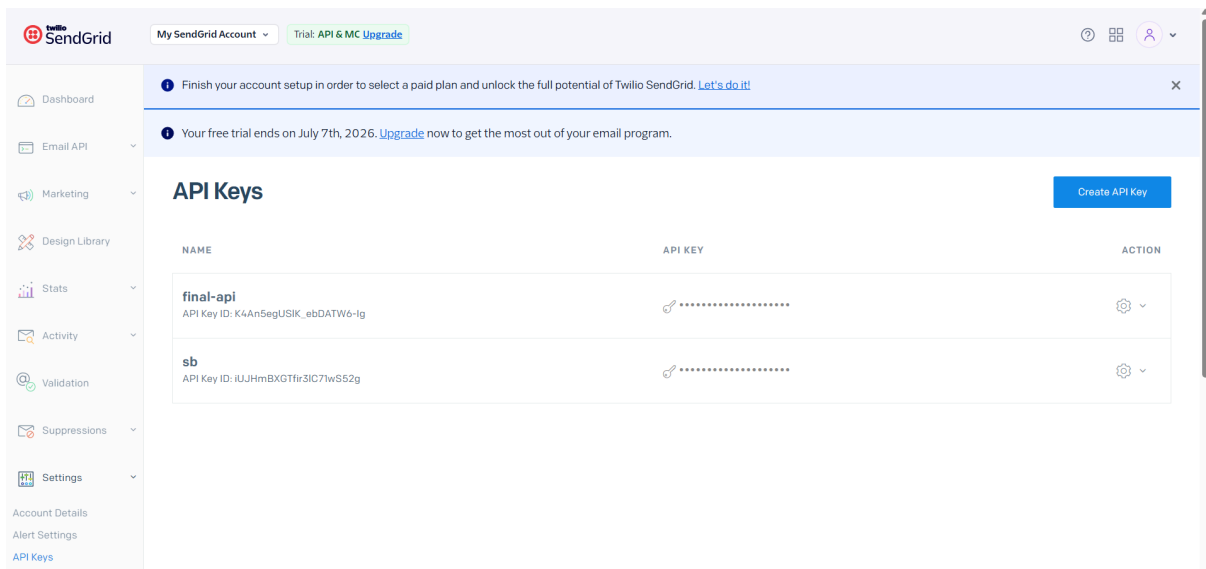
Save

Single Sender Verification ⓘ

Create New Sender

SENDERS	ADDRESS	NICKNAME	VERIFIED	ACTIONS
Siva Testing FROM sivajarvis5@gmail.com REPLY sivajarvis5@gmail.com	22/1 muthu irulappa street, ettayapuram Coimbatore, 635001 IND	Sivabalasankar		⋮
siva				

Generated SendGrid API Key



Added environment variables in .env.local

```
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ ls -la
total 280
drwxrwxr-x 14 siva-leadtap siva-leadtap 4096 May 9 14:27 .
drwxr-x--- 10 siva-leadtap siva-leadtap 4096 May 9 14:27 ..
-rw-rw-r-- 1 siva-leadtap siva-leadtap 127 May 8 17:10 .env.local
-rw-r--r-- 1 siva-leadtap siva-leadtap 12288 May 9 11:49 .env.local.swp
drwxrwxr-x 7 siva-leadtap siva-leadtap 4096 May 9 13:01 .git
drwxr-xr-x 3 siva-leadtap siva-leadtap 4096 May 9 13:51 .next
drwxrwxr-x 4 siva-leadtap siva-leadtap 4096 May 8 10:31 .npm
drwxrwxr-x 8 siva-leadtap siva-leadtap 4096 May 8 10:31 .nvm
drwxrwxr-x 5 siva-leadtap siva-leadtap 4096 May 9 05:55 .pm2
-rw-rw-r-- 1 siva-leadtap siva-leadtap 18 May 8 10:39 README.md
drwxrwxr-x 3 siva-leadtap siva-leadtap 4096 May 8 10:39 app
drwxrwxr-x 3 siva-leadtap siva-leadtap 4096 May 8 10:39 components
-rw-rw-r-- 1 siva-leadtap siva-leadtap 443 May 8 10:39 components.json
drwxrwxr-x 2 siva-leadtap siva-leadtap 4096 May 8 10:39 hooks
drwxrwxr-x 2 siva-leadtap siva-leadtap 4096 May 8 10:39 lib
-rw-rw-r-- 1 siva-leadtap siva-leadtap 262 May 9 11:39 next-env.d.ts
drwxrwxr-x 211 siva-leadtap siva-leadtap 12288 May 9 11:38 node_modules
-rw-rw-r-- 1 siva-leadtap siva-leadtap 173398 May 9 11:38 package-lock.json
-rw-rw-r-- 1 siva-leadtap siva-leadtap 2305 May 9 11:38 package.json
-rw-rw-r-- 1 siva-leadtap siva-leadtap 135 May 8 10:39 postcss.config.mjs
drwxrwxr-x 3 siva-leadtap siva-leadtap 4096 May 8 10:39 public
drwxrwxr-x 2 siva-leadtap siva-leadtap 4096 May 8 10:39 styles
-rw-rw-r-- 1 siva-leadtap siva-leadtap 2961 May 8 10:39 tailwind.config.ts
-rw-rw-r-- 1 siva-leadtap siva-leadtap 595 May 8 10:39 tsconfig.json
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ |
```

9. Application Build & PM2 Setup

9.1 Installed Dependencies

```
npm install

siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ npm install
up to date, audited 301 packages in 3s

52 packages are looking for funding
  run 'npm fund' for details

2 moderate severity vulnerabilities

To address all issues (including breaking changes), run:
  npm audit fix --force

Run 'npm audit' for details.

npm notice
npm notice New major version of npm available! 10.8.2 -> 11.14.1
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.14.1
npm notice To update run: npm install -g npm@11.14.1
npm notice
```

9.2 Built Next.js Application

```
npm run build
```

```
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ sudo npm run build
```

```
> my-v0-project@0.1.0 build
> next build
```

```
  ▲ Next.js 15.5.18
  - Environments: .env.local
```

```
Creating an optimized production build ...
```

```
✓ Compiled successfully in 13.8s
✓ Linting and checking validity of types
✓ Collecting page data
✓ Generating static pages (5/5)
✓ Collecting build traces
✓ Finalizing page optimization
```

Route (app)	Size	First Load JS
o /	66.2 kB	169 kB
o /_not-found	993 B	103 kB
f /api/sendgrid	123 B	103 kB
+ First Load JS shared by all	102 kB	
chunks/255-4f84124391a7dac4.js	46.2 kB	
chunks/4bd1b696-c023c6e3521b1417.js	54.2 kB	
other shared chunks (total)	1.94 kB	

```
o (Static) prerendered as static content
f (Dynamic) server-rendered on demand
```

9.3 Started Application with PM2

```
pm2 start npm --name "lt-nilavan" -- start
```

```
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ pm2 start npm --name "lt-nilavan" -- start
[PM2] Starting /usr/bin/npm in fork_mode (1 instance)
[PM2] Done.
```

id	name	namespace	version	mode	pid	uptime	o	status	cpu	mem	user	watching
0	lt-nilavan	default	N/A	fork	18321	0s	0	online	0%	25.7mb	siv...	disabled

9.4 Saved PM2 Processes

```
pm2 save
```

```
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ pm2 save
[PM2] Saving current process list...
[PM2] Successfully saved in /home/siva-leadtap/.pm2/dump.pm2
```

9.5 Enabled PM2 Auto Startup

```
pm2 startup
```

```
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ pm2 startup
[PM2] Init System found: systemd
[PM2] To setup the Startup Script, copy/paste the following command:
sudo env PATH=$PATH:/usr/bin /usr/lib/node_modules/pm2/bin/pm2 startup systemd -u siva-leadtap --hp /home/siva-leadtap

siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ sudo env PATH=$PATH:/usr/bin /usr/lib/node_modules/pm2/bin/pm2 startup systemd -u siva-leadtap --hp /home/siva-leadtap
[PM2] Init System found: systemd
Platform systemd
Template
[Unit]
Description=PM2 process manager
Documentation=https://pm2.keymetrics.io/
After=network.target

[Service]
Type=forking
User=siva-leadtap
LimitNOFILE=infinity
LimitNPROC=infinity
LimitCORE=infinity
Environment=PATH=/home/siva-leadtap/.nvm/versions/node/v20.2/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/usr/bin:/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin
Environment=PM2_HOME=/home/siva-leadtap/.pm2
PIDFile=/home/siva-leadtap/.pm2/pm2.pid
Restart=on-failure

ExecStart=/usr/lib/node_modules/pm2/bin/pm2 resurrect
ExecReload=/usr/lib/node_modules/pm2/bin/pm2 reload all
ExecStop=/usr/lib/node_modules/pm2/bin/pm2 kill
```

10. Nginx Reverse Proxy Configuration

10.1 Installed Nginx

```
sudo apt install nginx -y
```

```
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ sudo systemctl status nginx.service
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-05-09 11:46:24 UTC; 2h 49min ago
     Invocation: 64ad8c9881784247ab21de82062243ea
       Docs: man:nginx(8)
    Main PID: 11831 (nginx)
      Tasks: 3 (limit: 1741)
     Memory: 8.6M (peak: 30.4M)
        CPU: 229ms
    CGroup: /system.slice/nginx.service
            └─11831 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
              └─11832 "nginx: worker process"
                └─11833 "nginx: worker process"

May 09 11:46:24 ip-10-0-10-55 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
May 09 11:46:24 ip-10-0-10-55 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$
```

```
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ nginx -v
nginx version: nginx/1.28.3 (Ubuntu)
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$
```

10.2 Created Reverse Proxy Configuration

Configuration File

```
sudo nano /etc/nginx/sites-available/lt-nilavan
```

Nginx Configuration

```
server { listen 80; listen [::]:80;
```

```
server_name _;  
  
return 301 https://$host$request_uri;  
  
}
```

Enable Site

```
sudo ln -s /etc/nginx/sites-available/lt-nilavan /etc/nginx/sites-enabled/
```

Test Nginx

```
sudo nginx -t
```

```
siva-leadtap@ip-10-0-10-55:/etc/nginx/sites-available$ sudo nginx -t  
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok  
nginx: configuration file /etc/nginx/nginx.conf test is successful  
siva-leadtap@ip-10-0-10-55:/etc/nginx/sites-available$ |
```

Restart Nginx

```
sudo systemctl restart nginx
```

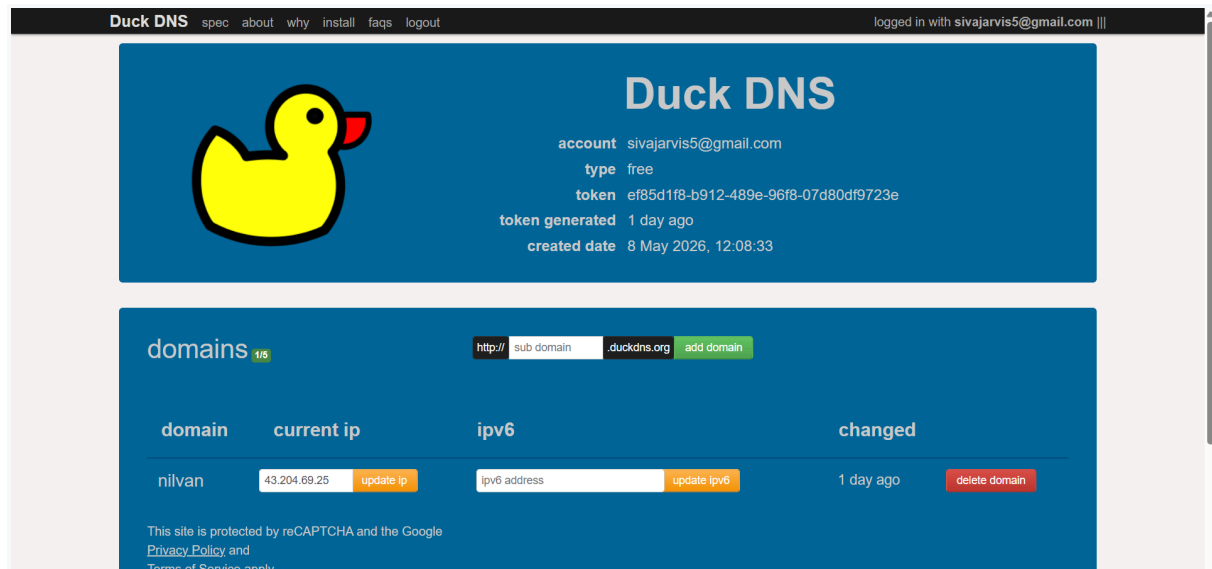
11. DuckDNS Configuration

11.1 Configured Domain

DuckDNS domain was mapped to the Elastic IP attached to the VPS.

Domain

```
https://nilvan.duckdns.org
```



```
nslookup nilvan.duckdns.org
```

```
siva-leadtap@ip-10-0-10-55:/etc/nginx/sites-available$ nslookup nilvan.duckdns.org
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   nilvan.duckdns.org
Address: 43.204.69.25

siva-leadtap@ip-10-0-10-55:/etc/nginx/sites-available$ |
```

12. SSL Configuration using Certbot

12.1 Installed Certbot

```
sudo apt install certbot python3-certbot-nginx -y
```

12.2 Generated SSL Certificate

```
sudo certbot --nginx -d nilvan.duckdns.org
```



```
siva-leadtap@ip-10-0-10-55:/etc/nginx/sites-available$ curl -I https://nilvan.duckdns.org
HTTP/2 200
server: nginx
date: Sat, 09 May 2026 14:40:36 GMT
content-type: text/html; charset=utf-8
content-length: 47835
vary: rsc, next-router-state-tree, next-router-prefetch, next-router-segment-prefetch, Accept-Encoding
x-nextjs-cache: HIT
x-nextjs-prerender: 1
x-nextjs-prerender: 1
x-nextjs-stale-time: 300
x-powered-by: Next.js
cache-control: s-maxage=31536000
etag: "xdmb4g0t10wo"
x-frame-options: DENY
x-content-type-options: nosniff
referrer-policy: strict-origin-when-cross-origin
strict-transport-security: max-age=63072000; includeSubDomains; preload
permissions-policy: geolocation=(), microphone=(), camera=()
content-security-policy: default-src 'self'; base-uri 'self'; object-src 'none'; frame-ancestors 'none'; form-action 'self'; script-src 'self' 'unsafe-inline' 'unsafe-eval' https;; style-src 'self' 'unsafe-inline' https;; img-src 'self' data: blob: https;; font-src 'self' data: https;; connect-src 'self' https;; frame-src 'self' https;;
```

```
siva-leadtap@ip-10-0-10-55:/etc/nginx/sites-available$ sudo certbot certificates
Saving debug log to /var/log/letsencrypt/letsencrypt.log

-----
Found the following certs:
Certificate Name: nilvan.duckdns.org
Serial Number: 56e5d714bac06e44df73b01bbcd27943279
Key Type: ECDSA
Domains: nilvan.duckdns.org
Expiry Date: 2026-08-06 12:03:07+00:00 (VALID: 88 days)
Certificate Path: /etc/letsencrypt/live/nilvan.duckdns.org/fullchain.pem
Private Key Path: /etc/letsencrypt/live/nilvan.duckdns.org/privkey.pem
-----
siva-leadtap@ip-10-0-10-55:/etc/nginx/sites-available$ |
```

13. HTTPS Nginx Configuration

13.1 Added SSL Configuration

HTTPS Nginx Configuration

```
server {
    listen 443 ssl;
    listen [::]:443 ssl;

    http2 on;

    server_name nilvan.duckdns.org;

    # SSL
    ssl_certificate /etc/letsencrypt/live/nilvan.duckdns.org/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/nilvan.duckdns.org/privkey.pem;

    ssl_protocols TLSv1.2 TLSv1.3;
    ssl_prefer_server_ciphers on;

    # Security headers
    add_header X-Frame-Options "DENY" always;
    add_header X-Content-Type-Options "nosniff" always;
    add_header Referrer-Policy "strict-origin-when-cross-origin" always;
    add_header Strict-Transport-Security "max-age=63072000;
includeSubDomains; preload" always;
```

```

    add_header Permissions-Policy "geolocation=(), microphone=(), camera=()"
always;
    add_header Content-Security-Policy "default-src 'self'; base-uri 'self';
object-src 'none'; frame-ancestors 'none'; form-action 'self'; script-src
'self' 'unsafe-inline' 'unsafe-eval' https;; style-src 'self' 'unsafe-inline'
https;; img-src 'self' data: blob: https;; font-src 'self' data: https;;
connect-src 'self' https;; frame-src 'self' https;;" always;

# Block hidden files
location ~ /\.(!well-known).* {
    deny all;
    access_log off;
    log_not_found off;
}

# Block .git access
location /.git {
    deny all;
    return 403;
}

# API rate limiting
location /api/ {
    limit_req zone=api_limit burst=1 nodelay;

    proxy_pass http://127.0.0.1:3000;

    proxy_http_version 1.1;

    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto https;

    # Mitigate Next.js middleware vulnerability
    proxy_set_header x-middleware-subrequest "";

    proxy_cache_bypass $http_upgrade;
}

# Main app
location / {
    proxy_pass http://127.0.0.1:3000;

    proxy_http_version 1.1;

    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection "upgrade";

    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto https;

```

```
proxy_set_header x-middleware-subrequest "";
```

```
proxy_cache_bypass $http_upgrade;
```

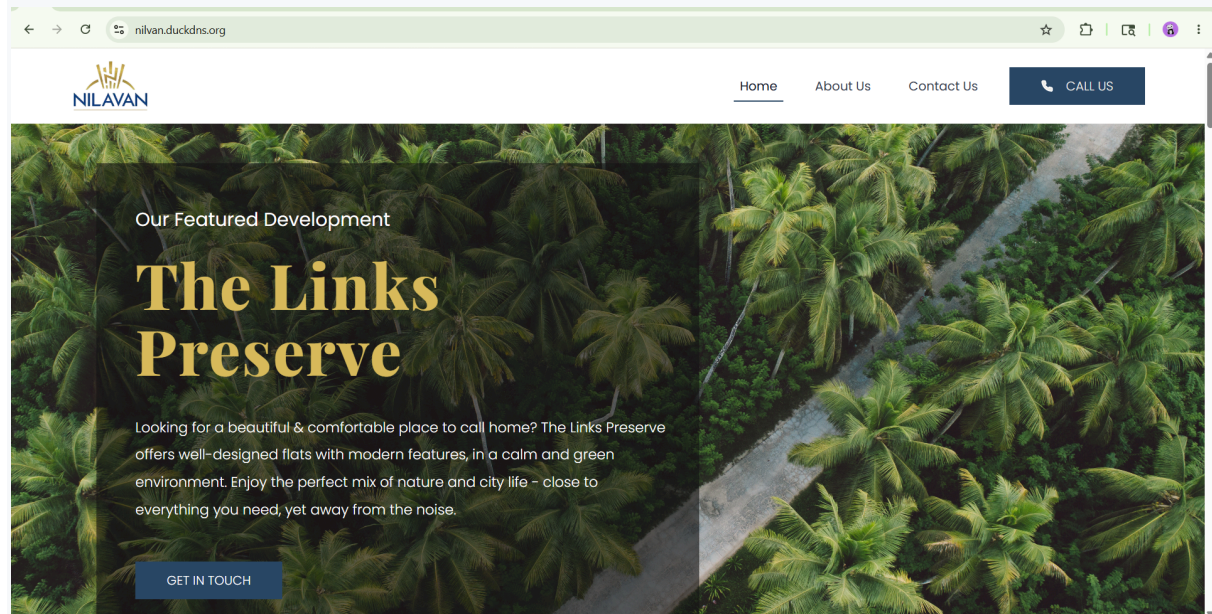
```
}
```

```
}
```

Restart Nginx

```
sudo nginx -t
```

```
sudo systemctl restart nginx
```



14. Security Improvements

Implemented Security Controls

Security Feature	Status
Non-root deployment user	Enabled
SSH Port Changed	Enabled
Root Login Disabled	Enabled
Password Authentication Disabled	Enabled
Public Key Authentication	Enabled
HTTPS Enabled	Enabled
Reverse Proxy Configured	Enabled
PM2 Auto Start	Enabled
.git Public Access Blocked	Enabled
Elastic IP Configured	Enabled

15. Final Deployment Verification

Production URL

<https://nilvan.duckdns.org>

Website Verification

Check Local App

```
curl -I http://localhost:3000
```

```
ubuntu@ip-10-0-10-55:~$ curl -I http://localhost:3000
HTTP/1.1 200 OK
Vary: rsc, next-router-state-tree, next-router-prefetch, next-router-segment-prefetch, Accept-Encoding
x-nextjs-cache: HIT
x-nextjs-prerender: 1
x-nextjs-prerender: 1
x-nextjs-stale-time: 300
X-Powered-By: Next.js
Cache-Control: s-maxage=31536000
ETag: "xdmb4g0t10wo"
Content-Type: text/html; charset=utf-8
Content-Length: 47835
Date: Sun, 10 May 2026 04:30:30 GMT
Connection: keep-alive
Keep-Alive: timeout=5
```

Check HTTPS Response

```
curl -I https://nilvan.duckdns.org
```

```
ubuntu@ip-10-0-10-55:~$ curl -I https://nilvan.duckdns.org
HTTP/2 200
server: nginx
date: Sun, 10 May 2026 04:30:50 GMT
content-type: text/html; charset=utf-8
content-length: 47835
vary: rsc, next-router-state-tree, next-router-prefetch, next-router-segment-prefetch, Accept-Encoding
x-nextjs-cache: HIT
x-nextjs-prerender: 1
x-nextjs-prerender: 1
x-nextjs-stale-time: 300
x-powered-by: Next.js
cache-control: s-maxage=31536000
etag: "xdmb4g0t10wo"
x-frame-options: DENY
x-content-type-options: nosniff
referrer-policy: strict-origin-when-cross-origin
strict-transport-security: max-age=63072000; includeSubDomains; preload
permissions-policy: geolocation=(), microphone=(), camera=()
content-security-policy: default-src 'self'; base-uri 'self'; object-src 'none'; frame-ancestors 'none'; form-action 'self'; script-src 'self' 'unsafe-inline' 'unsafe-eval' https; style-src 'self' 'unsafe-inline' https; img-src 'self' data blob: https; font-src 'self' data: https; connect-src 'self' https; frame-src 'self' https;
```

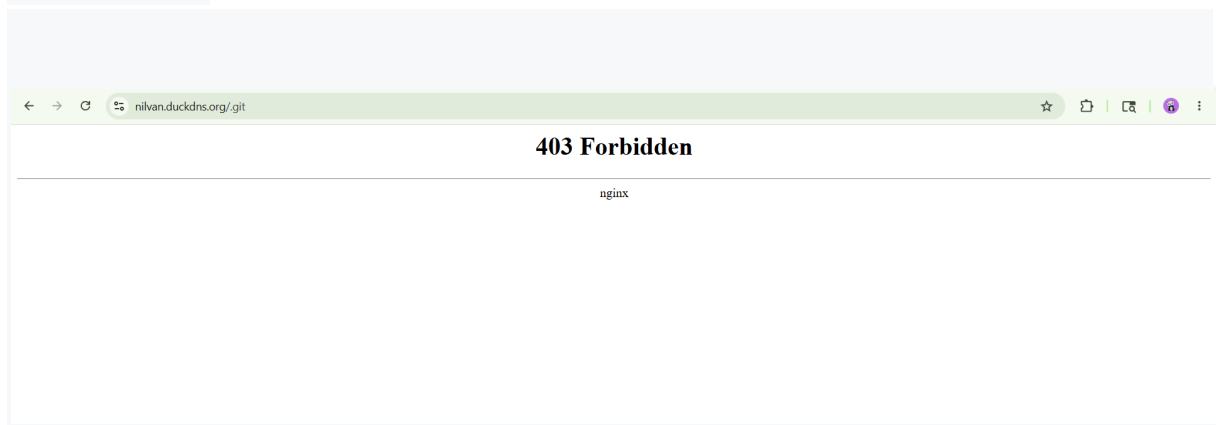
Verify .git Access Blocked

```
curl -I https://nilvan.duckdns.org/.git
```

```
ubuntu@ip-10-0-10-55:~$ curl -I https://nilvan.duckdns.org/.git
HTTP/2 403
server: nginx
date: Sun, 10 May 2026 04:31:07 GMT
content-type: text/html
content-length: 146
x-frame-options: DENY
x-content-type-options: nosniff
referrer-policy: strict-origin-when-cross-origin
strict-transport-security: max-age=63072000; includeSubDomains; preload
permissions-policy: geolocation=(), microphone=(), camera=()
content-security-policy: default-src 'self'; base-uri 'self'; object-src 'none'; frame-ancestors 'none'; form-action 'self'; script-src 'self' 'unsafe-inline' 'unsafe-eval' https;; style-src 'self' 'unsafe-inline' https;; img-src 'self' data: blob: https;; font-src 'self' data: https;; connect-src 'self' https;; frame-src 'self' https;;
```

Expected Output:

403 Forbidden



17. Conclusion

The Next.js application was successfully deployed on Ubuntu 24.04 with:

- Secure SSH hardening
- Non-root deployment user
- Elastic IP configuration
- GitHub SSH deployment
- PM2 process management
- Nginx reverse proxy
- SSL/TLS using Let's Encrypt
- DuckDNS domain configuration
- Production-ready hardened environment

The deployment follows modern security best practices suitable for production environments